

Thermal, viscoelastic and contact verification toward gap-constrained annealing of FFF-printed PLA

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Abstract

Quantified fixture clearance is a potential means of controlling deformation during annealing of fused-filament-fabricated polylactic acid, but prediction requires compatible material data and verified thermal, mechanical and contact implementations. This study develops and numerically verifies components of such a framework using Ansys Mechanical APDL 2026 R1. A formulation-specific, 23-branch generalized-Maxwell reference is evaluated alongside separate thermal, structural and contact benchmarks; these do not constitute a coupled annealing simulation. The transient plane-wall comparison gives a maximum sampled center-temperature error of $0.005180700\text{ }^{\circ}\text{C}$ and maximum excursion-relative error of 0.157660518% . Six structural/contact reference cases pass 114 scalar comparisons; the isothermal Prony calculation has maximum stress error $3.8271 \times 10^{-6}\text{ MPa}$. Refinement from a 90×12 to a 120×16 structural grid changes benchmark vertical warpage by 0.9327% and interior residual stress by 2.0266% . A 40-to-60-element normal-Lagrange contact refinement changes mean pressure by 0.12698% , with zero reported penetration. These results verify declared numerical configurations, not production annealing behavior. Compatible thermal functions, bulk irreversible strain, a non-isothermal ANSYS adapter and independent physical validation remain incomplete. Consequently no production clearance benefit, processing optimum or uncertainty interval is established. The contribution is a reproducible verification foundation and an explicit account of the evidence still required for predictive constrained-annealing analysis.

Keywords: fused filament fabrication; polylactic acid; annealing; fixture clearance; numerical verification; viscoelasticity; contact

1. Introduction

Thermal post-processing of FFF-printed PLA can alter dimensions and mechanical response [1,2,27,28]. Geometric support introduces an additional constraint: suppressed deformation during a hot hold need not persist after cooling and release. Predicting released dimensions therefore requires the actual thermal history, history-dependent material response and evolving contact, with validation specific to the observable being claimed.

The primary research question is how temperature, attained-temperature holding time and initial fixture clearance influence transient exposure, irreversible dimensional response, warpage and residual stress relative to free annealing. A rectangular coupon and opposed-plate fixture define the intended comparison. This article reports the numerical verification achieved so far and distinguishes it from production predictions that the available material and validation evidence cannot yet support.

The objectives are to establish traceable material and geometry definitions, verify component calculations, quantify discretization effects, assess independent validation readiness and delimit defensible sensitivity and decision analysis. The achieved contribution is component verification with immutable solver evidence. Prediction of a clearance–distortion–stress trade-off remains an unachieved objective. No first-use claim is made for supported annealing, ANSYS, Prony series or optimization.

2. Background and research gap

Supported annealing is established. Powder-mould experiments [11], treatment in different media [12], mould-supported processing [26], salt-assisted remelting [25] and tough-PLA encapsulation [24] preclude novelty claims based solely on support. High-heat PLA conditioning [13], porosity/interlayer studies [29] and crystallization/bonding competition [30] show why grades and print architectures must not be pooled as interchangeable evidence. Remelting is outside the proposed sub-melting domain.

Wijnen et al. [6] modeled deformation and examined annealing dimensions; Trofimov et al. [7] compared thermal and distortion predictions with experiments. Bute et al. [5] separated irreversible directional strain from subsequent-cycle expansion and related recovery to ANSYS printing stresses. Issabayeva and Shishkovsky [9] used viscoelastic representations, while Chapuis et al. [8] developed a pre-strain/Maxwell laminate model for thermally activated shape change. Temperature shifting [31], relaxation measurements [22] and viscoelastic–viscoplastic characterization [21] support history dependence but do not independently validate a bulk fixture-constrained coupon.

Orthotropic FFF mechanics [16], thermal/deformation calculations [10], composite crystallization modeling [15], thermal mesh adaptation [32] and finite-difference thermal studies [37] establish relevant component methods. Virgin/processed PLA crystallization [17], nucleation [18], filament kinetics [19], deformation-assisted annealing [20], phase-field mechanics [33] and injection-moulded grade comparisons [34] provide context rather than transferable coefficients. Constitutive calibration must retain formulation, initial state, architecture and temperature-domain compatibility.

Annealing optimization [23], multiscale FE/desirability analysis [14] and physics-informed recycled-PLA optimization [35] precede this work. Derringer–Suich desirability [4] is established methodology, not evidence of physical validity. ASTM F3489-23 [3] supports appraisal of material-extrusion mechanical data; it does not certify a simulation or supply an annealing law.

The original targeted review covers verified records through 12 September 2026. A focused audit search on 30 September 2026 additionally identified an explicit mould-clearance precedent: Hussam et al. [41, Section 2.2 and Fig. 4] compare 1.0 mm and 0.6 mm clearances during ABS annealing. Thus even quantified annealing-mould clearance is not novel by itself. ABS observations do not supply PLA parameters or validate the proposed fixture. The focused update is not an exhaustive systematic review. In the accessible evidence, the reviewed sources did not establish the entire chain of quantified annealing-fixture clearance, evolving contact, irreversible directional response and independently supported released geometry/stress with uncertainty. This is a bounded search finding, not proof of absence.

Supplementary Table S1 compares the ten closest studies. The proposed gap remains an evidence requirement, not an achieved predictive capability.

3. Computational methodology

3.1. Research framework

The workflow separates material characterization, numerical verification, discretization assessment, independent physical validation and production inference. Actual field solutions in this article are reference problems. The full coupon/fixture annealing model is not executable with the present evidence, and no production campaign was run. Missing values remain unavailable rather than zero.

Mechanical APDL reports release 2026 R1, build 26.1, update 20260202 with the Ansys Mechanical Enterprise Academic Student product. Workbench and Mechanical executables report 26.1. The batch workflow uses explicit case files and input/output hashes. Student limits reported for this release are 128,000 nodes/elements and up to four HPC cores [40]. The host has eight physical cores, 16 logical processors and 15.82 GiB RAM. PyMechanical is not used; installed DesignXplorerer/optiSLang were not exercised. Claims are limited to the capabilities actually run.

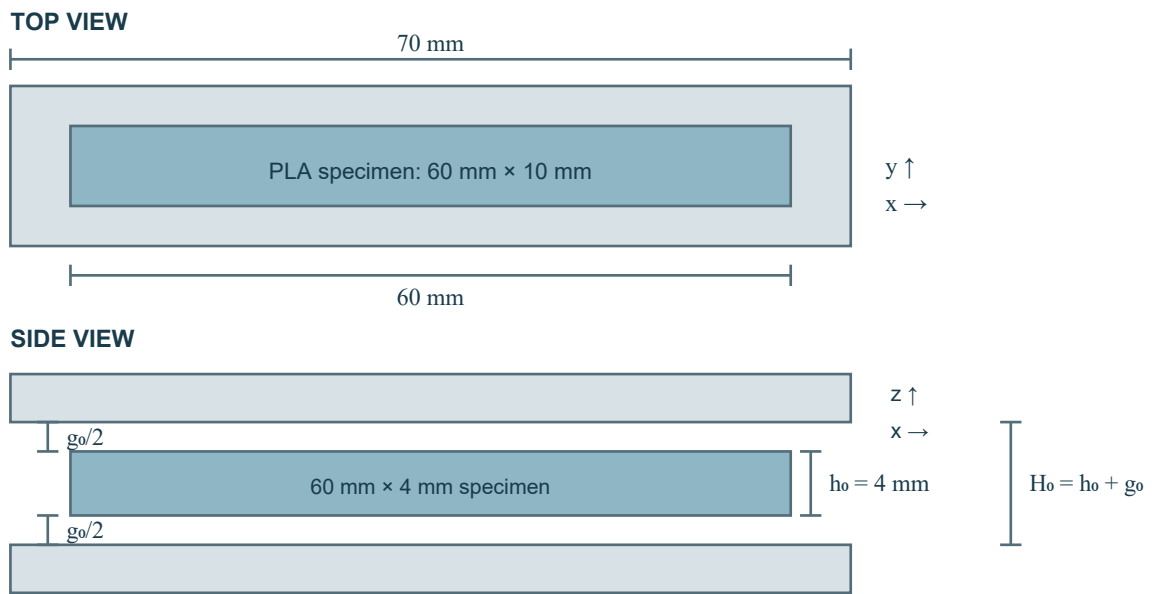
3.2. Geometry

The intended solid coupon is 60 mm × 10 mm × 4 mm, adopting only the regular-plate dimensions in Trofimov et al. [7, Fig. 4(a)]. Their Raise3D properties and validation outcomes are not transferred to Prusament PLA. Two candidate AISI 304 plates each measure 70 mm × 20 mm × 5 mm, leaving a 5 mm plan margin on each side. A geometry-only Mechanical APDL build created three volumes; it contains no element type, mesh, material, load, analysis type or solve command.

The total initial clearance and normalized clearance are defined at a common reference temperature by

$$g_0 = H_0 - h_0, \quad \gamma = g_0/h_0. \quad (1)$$

Here H_0 is the inside plate spacing and h_0 is specimen thickness. The initially centered coupon has half the clearance above and below. Geometry screening uses $\gamma = 0, 0.0025, 0.005, 0.01$ and 0.02 , corresponding to $g_0 = 0, 0.01, 0.02, 0.04$ and 0.08 mm. These are design choices, not approved treatment levels or predictions of contact onset. FREE omits the fixture; it is not the zero-gap contact condition.



Each fixture plate: 70 mm × 20 mm × 5 mm; 5 mm plan margin per side
Figure 1. Candidate coupon and opposed-plate geometry at the reference state. The schematic contains no solved mesh, temperature, contact or deformation field.

3.3. Material properties

Prusament PLA is the formulation-specific constitutive reference. Chapuis et al. [8, supplementary Tables B.1–B.2] provide $E_{\infty} = 10.598 \text{ MPa}$, $\nu = 0.35$, $\alpha = 68.0 \text{ } \mu\text{m m}^{-1} \text{ K}^{-1}$, $T_g = 65 \text{ }^{\circ}\text{C}$ and 23 branch pairs. Direct characterization spans 23–85 $^{\circ}\text{C}$. The source-calculated instantaneous modulus is 1691.594 MPa. Supplementary Table S2 retains the full spectrum. Supplier density of 1240 kg m^{-3} has no reported test temperature [36] and remains screening information, not an admitted production law.

Compatible temperature-dependent Prusament functions remain absent for $k(T)$ and $c_p(T)$. A complete orthotropic property set, initial printed stress, bulk recovery law and crystallization parameters are absent. Other formulations [5,16,22,37] remain comparators. Supplier modulus intervals and same-source bilayer identification brackets are not bulk annealing distributions. No cross-grade uncertainty envelope is assigned.

AISI 304 is the candidate fixture material [38, Table 1]. Its reported density, heat capacity, conductivity, expansion, modulus and Poisson ratio are tabulated at 20 $^{\circ}\text{C}$, 100 $^{\circ}\text{C}$ and 200 $^{\circ}\text{C}$ (Supplementary Table S3). Implemented interpolation is

$$p(T) = p_a + (p_b - p_a)(T - T_a)/(T_b - T_a). \quad (2)$$

The code reproduces knots and rejects extrapolation. The expansion table's tangent-versus-mean convention and reference temperature remain unresolved; interpolation therefore does not authorize fixture thermal strain.

3.4. Constitutive model

The executable material-point core is small-strain isotropic linear viscoelasticity with constant Poisson ratio. Its strain partition is

$$\begin{aligned} \varepsilon &= \varepsilon^e + \varepsilon^{ve} + \varepsilon^{th}, \\ \varepsilon^m &= \varepsilon - \varepsilon^{th}. \end{aligned} \quad (3)$$

This matches the update routine. Effective elastic strain is instantaneous compliance applied to stress; delayed strain is mechanical strain minus that elastic contribution. No bulk irreversible-strain evolution term is implemented. Its absence is a limitation, not a physical zero assigned to production PLA. The separate imposed-eigenstrain benchmark does not supply such a law.

Reversible expansion and isotropic stiffness are

$$\begin{aligned} \varepsilon^{th}(T) &= \alpha(T - T_r)I, \\ \alpha &= 68.0 \times 10^{-6} \text{ K}^{-1}. \end{aligned} \quad (4)$$

$$\begin{aligned} C(E, \nu) : A &= 2G A + \lambda \text{tr}(A)I, \\ G &= E/[2(1 + \nu)], \quad \lambda = E\nu/[(1 + \nu)(1 - 2\nu)]. \end{aligned} \quad (5)$$

I is the identity tensor, T_r the reference temperature, A a symmetric strain tensor, G the shear modulus and λ the Lamé parameter. The code uses tensor shear; engineering-shear interfaces require conversion. The three-dimensional isotropic extension of the source laminate model is an assumption, not measured FFF isotropy.

In reduced time ξ , the network, Prony normalization and isothermal relaxation modulus are

$$\sigma = C_{\infty} : \varepsilon^m + \sum_{i=1}^{23} s_i, \\ \partial s_i / \partial \xi + s_i / \tau_i = C_i : \partial \varepsilon^m / \partial \xi. \quad (6)$$

$$E_{\text{inst}} = E_{\infty} + \sum_{i=1}^{23} k_i, \quad g_i = k_i / E_{\text{inst}}, \\ G_{\text{inst}} = E_{\text{inst}} / [2(1 + \nu)], \quad K_{\text{inst}} = E_{\text{inst}} / [3(1 - 2\nu)]. \quad (7)$$

$$E_r(t, T) = E_{\infty} + \sum_{i=1}^{23} k_i \exp[-t / (a_T \tau_i)]. \quad (8)$$

Branch stresses are s_i , relaxation times τ_i and source tensile moduli k_i . Equal fractional shear and bulk spectra preserve constant ν . E_r is the isothermal relaxation modulus. Temperature changes the clock; no separate modulus multiplier is added. Native ANSYS spectra were checked only at 65 °C [39], without shifting.

The source piecewise clock and reduced time are

$$\log_{10} a_T = C_3 (1/T_K - 1/T_{g,K}), \quad T < T_g; \\ \log_{10} a_T = -C_1 (T - T_g) / [C_2 + (T - T_g)], \quad T \geq T_g. \quad (9)$$

$$\xi(t) = \int_0^t [a_T(T(s))]^{-1} ds. \quad (10)$$

$C_1 = 17.4$, $C_2 = 51.6$ K and $C_3 = 35\,000$ K [8]; T_K and $T_{g,K}$ are kelvin temperatures. The base-10 shift multiplies reference relaxation times. The code rejects values outside 23–85 °C and splits ramps at 65 °C. Composite Simpson quadrature integrates the clock for a linear temperature path; this utility is distinct from the frozen-temperature substep update.

For strain linear in reduced time, the implemented update is

$$r_i = \Delta \xi / \tau_i, \quad A_i = \exp(-r_i), \\ B_i = [1 - \exp(-r_i)] / r_i, \\ s_i^{n+1} = A_i s_i^n + B_i C_i : \Delta \varepsilon^m. \quad (11)$$

The exponential difference is evaluated accurately near zero. The physical-time update uses midpoint temperature to approximate reduced-time increments; it is exact for an isothermal linear-strain segment and requires refinement for a changing temperature. No verified full non-isothermal ANSYS adapter exists. Positive moduli and times support the isothermal reference, not validity under arbitrary strain, duration or annealing history. The very short fitted relaxation times describe a source master-curve representation; this study neither independently resolves nor assigns molecular meaning to every branch. ν and α are constant source assumptions, and equal bulk/shear relaxation fractions are not a measurement of volumetric relaxation.

3.5. Thermal formulation

The executed transient conduction reference solves

$$\rho c_p \partial T / \partial t = \nabla \cdot (k \nabla T). \quad (12)$$

Here ρ , c_p and k are density, heat capacity and conductivity. The verified problem uses constant numerical coefficients and convection, specified in Section 3.9. Radiation is omitted from this benchmark to retain an

analytical comparison. Production still needs compatible thermal functions, physical convection, a radiation decision and thermal-contact conductance; no absent coefficient is replaced by zero.

3.6. Mechanical formulation

The reference cases solve quasi-static force balance without body force,

$$\nabla \cdot \sigma = 0. \quad (13)$$

The uniform SOLID185 tests and plane-stress cantilever use small-deformation kinematics. The separate plane-strain surface-contact model uses NLGEOM,ON; its equilibrium is evaluated in the current configuration. It is a geometrically nonlinear elastic contact benchmark, not the small-strain Prusament production model. Uniform expansion, imposed eigenstrain and isothermal memory are isolated before contact is introduced. Production extraction would require fixed landmarks, aligned signed dimensions and a declared cooled/released state. The three-dimensional dimensional-error and best-fit-plane extractors are not implemented and are not presented as executed equations. The convergence benchmark instead measures the maximum absolute vertical deviation from an ordinary least-squares line through the deformed two-dimensional top edge at 62 s. $W_{\max}$ denotes this coordinate-dependent benchmark measure, not perpendicular distance or a rigid-rotation-invariant surface-flatness measure.

3.7. Annealing strain

No bulk annealing-induced irreversible-strain relation is parameterized. Prusament programmed pre-strains refer to thin bilayers [8]; another grade's directional strain [5] is not transferable. Explicit evidence-gap errors prevent production use of absent laws. Initial stress and a recovery field cannot be combined without checking identifiability and double-counting.

No crystallization kinetic equation is included. Compatible initial state and non-isothermal kinetics are lacking [17–20]; an isothermal Avrami law is not evaluated instantaneously along a temperature ramp. Reversible expansion, finite-time delayed deformation and irreversible distortion remain distinct.

3.8. Fixture and contact

Ideal unilateral normal contact satisfies

$$g_n \geq 0, \quad p_n \geq 0, \quad g_n p_n = 0. \quad (14)$$

Positive g_n denotes separation and p_n compressive pressure. CONTA178 node-to-stop and two-dimensional CONTA172/TARGE169 models verify limited forms of this relation. The latter uses a 10 mm × 4 mm plane-strain elastic block, a 0.020 mm rigid-stop gap, $E = 2000$ MPa, $\nu = 0.3$ and $\alpha = 10^{-4} \text{ K}^{-1}$, heated uniformly from 20 °C to 120 °C. These are synthetic reference inputs. Its lower edge has zero vertical displacement and only its lower-left node zero horizontal displacement; this is not a traction-free coupon. The pressure mean is the arithmetic mean of nonnegative exported element pressures, including zero entries, rather than reaction divided by a measured current contact area. Peak pressure is the largest exported element value. Reaction has units N mm^{-1} of out-of-plane depth. A positive element-average pressure is only an engagement screen; it does not prove closure at every integration point. Penalty enforcement permits numerical penetration; normal-Lagrange comparison supplies the reported zero-penetration solution. Production three-dimensional fixture contact, friction, spacers and heat transfer are incomplete. A stiff plate is neither infinitely rigid nor thermally inert.

3.9. Boundary conditions

The thermal reference is a 1 mm × 1 mm × 10 mm wall with adiabatic side faces and equal convection on its two z faces. Numerical fixtures are $k = 0.5 \text{ W m}^{-1} \text{ K}^{-1}$, $c_p = 1000 \text{ J kg}^{-1} \text{ K}^{-1}$, $\rho = 1000 \text{ kg m}^{-3}$ and $h = 5 \text{ W m}^{-2} \text{ K}^{-1}$. The body starts at 20 °C. Ambient temperature rises to 80 °C during 0–300 s, holds through 900 s, falls to 20 °C through 1200 s and remains there through 1800 s. Half-thickness is 0.005 m and $Bi = 0.05$. These are numerical verification inputs, not a measured PLA cycle.

Intended FREE surfaces are traction-free with verified rigid-body stabilization. GAP is initially centered without preload. Gravity is omitted as an isolation assumption; symmetry must not suppress investigated warpage. Production convection, hold attainment, cooling, release and observation time remain unspecified. Same-oven and matched-part-temperature comparisons are distinct and neither exists as a solved production pair.

3.10. Mesh

The thermal reference uses SOLID70; uniform structural tests use one eight-node SOLID185 brick of 10 mm × 1 mm × 1 mm. Convergence is studied separately on a plane wall, a 60 mm × 4 mm two-layer Prony cantilever of unit thickness and a 10 mm × 4 mm plane-strain contact block with 0.020 mm stop gap. The 65 °C cantilever has an upper-layer relaxation-time multiplier of 10, a numerical fixture rather than a PLA property.

Structural refinement measures vertical fitted-line warpage, maximum top-edge displacement magnitude and the 95th percentile of interior element-centroid von Mises stress over $6 \text{ mm} \leq x \leq 54 \text{ mm}$ at 62 s. The cantilever remains clamped; these are finite-time post-unloading responses, not fully released or permanent residuals. Thermal lag is the 80 °C ambient setpoint minus center temperature at 300 s. The reported thermal secant gradient is the absolute difference between mean surface and center temperatures divided by 5 mm at that instant; it is not the maximum local field gradient. The profile measure is the root-mean-square temperature excursion from 20 °C over archived nodes at 300 s, 900 s, 1200 s and 1800 s. Contact refinement measures exported element-average pressures, reaction and penetration. Adjacent relative change is

$$\delta_q = |q_{\text{fine}} - q_{\text{medium}}| / |q_{\text{fine}}| \times 100\%. \quad (15)$$

A zero denominator leaves percentage change undefined. Two zero values establish only zero absolute change. The historical convergence CSV uses a zero-percent sentinel for zero-to-zero penetration; the corrected audit table leaves those percentage cells blank and records the absolute comparison separately. Predeclared limits are 2% for global mesh responses and 5% for local stress/peak pressure. These limits are engineering screening choices, not formal error bounds. Adjacent changes do not prove an asymptotic convergence rate or an extrapolated exact solution. Mesh and time are refined separately, so their interaction is not bounded. The full matrices remain archived. No production mesh is selected from these differently configured problems.

3.11. Time stepping

The thermal time-step study compares 5 s, 1 s, 0.25 s and 0.125 s on a 64-element wall. Structural refinement compares ramp/hold pairs 0.1/1 s, 0.05/0.5 s, 0.01/0.1 s and 0.005/0.05 s on a fixed 60 × 8 grid. The −0.001 N cantilever load rises over 0–1 s, holds through 31 s, unloads through 32 s and remains zero through 62 s. Limits are 1% for global time-step responses and 3% for local stress. The uniform Prony patch test needs finer ramp increments; settings are not transferred between problems.

3.12. Verification

The plane-wall center reference uses step response and Duhamel superposition,

$$S_c(t) = 1 - \sum_{n=0}^{\infty} A_n \exp(-\zeta_n^2 \alpha t / L^2), \quad \zeta_n \tan \zeta_n = Bi, \\ A_n = 4 \sin \zeta_n / (2\zeta_n + \sin 2\zeta_n), \quad T_c(t) = T_i + \int_0^t S_c(t - \tau) dT_{\infty}(\tau). \quad (16)$$

In this equation $\alpha = k/(\rho c_p)$ is thermal diffusivity, distinct from the structural expansion coefficient; L is half-thickness. The implementation uses 200 roots and analytical integration over each linear ambient segment. The accepted comparison uses 640 elements, 1025 nodes, 40 elements through thickness and 0.25 s steps. Limits are 0.05 °C absolute error and 0.25% relative to the reference excursion from 20 °C; kelvin-based error is also retained.

Elastic/eigenstrain checks use $E = 2000$ MPa, $\nu = 0.3$ and $\alpha = 10^{-5} \text{ K}^{-1}$ as fixtures. Transverse symmetry permits homogeneous Poisson response; the $x = 0$ face is restrained axially and the opposing face is coupled in x . Temperature cycles from 20 °C to 30 °C to 80 °C and back with ten substeps per ramp. For uniform free strain e , $L = 10$ mm and $A = 1 \text{ mm}^2$, references are

$$u_x = Le, \sigma_x = 0 \text{ (free)}; \quad u_x = 0, \sigma_x = -Ee \text{ (fixed)}; \\ R_x = -A\sigma_x, \quad \varepsilon_y = \varepsilon_z = e - \nu\sigma_x/E. \quad (17)$$

Synthetic retained-eigenstrain tests add -0.001 isotropic strain through equivalent initialized stress, then compare fixed, free and released states. This is not a calibrated annealing law. Node contact adds a frictionless rigid stop at a 0.002 mm gap, with penetration/force tolerances 10^{-9} mm and 10^{-9} N. Reaction, displacement, force, status and opening are checked independently.

The Prony test uses all branches at 65 °C with the same thermal reference, avoiding shifting. Axial strain rises to 0.001 over 0–1 s, holds through 31 s, returns to zero through 32 s and remains zero through 62 s. A ramp on $[a,b]$ at strain rate v contributes, for $q = \min(t,b) > a$,

$$\sigma(t) = E_{\infty} v(q - a) + \sum_{i=1}^{23} E_i v \tau_i [\exp(-(t - q)/\tau_i) - \exp(-(t - a)/\tau_i)]. \quad (18)$$

E_i denotes the branch modulus k_i ; loading and unloading contributions are summed. Accepted increments are 0.0001 s on ramps and 0.1 s on holds. Each scalar error must be no greater than 10^{-7} in its unit plus 0.1% of the reference magnitude. Zero references have undefined relative error. Held-zero-strain stress after unloading measures memory, not free recovery.

3.13. Independent validation

The primary reserved dataset is Lluch-Cerezo et al. [11, Table 5]: six mould-free conditions, each with signed length, width and height means from five specimens. Ultimaker Pearl White PLA bars measure 80 mm $\times$ 10 mm $\times$ 4 mm, with longitudinal roads, full infill and 0.2 mm layers; the furnace ramp is 10 °C min⁻¹ and treatment duration 120 min. Actual mould-free support, specimen/cooling histories and observation delay remain unresolved. Supplementary Table S4 preserves the 18 literature means.

Chapuis et al. [8] supply calibration and are excluded from independent validation. External targets have not guided fitting; reservation is prospective but not blinded. Tuning to a condition would reclassify its correlated directions as calibration. Three selected maxima from Stojković et al. [2] remain quarantined owing to sign, selection and measurement limitations. Powder-supported or remelting cases cannot validate the plate fixture.

For matched prediction p_i and observation y_i , implemented metrics are

$$e_i = p_i - y_i, a_i = |e_i|, r_i = 100a_i/|y_i| (y_i \neq 0). \quad (19)$$

$$\text{MAE} = (1/n) \sum_i a_i, \text{RMSE} = \sqrt{[(1/n) \sum_i e_i^2]}. \quad (20)$$

$$a_i \leq b_{\text{num}, i} + b_{\text{meas}, i}. \quad (21)$$

The inequality is a deterministic discrepancy screen using independent numerical and measurement bounds, not a confidence interval. Missing bounds make classification indeterminate. Directional percentage-change errors use percentage points. Responses and grades are not pooled, and R^2 is not used for sparse validation. Metric functions are tested but no matched ANSYS prediction exists.

3.14. Parametric design

Candidate temperatures are 80 °C, 95 °C and 110 °C, with holds of 30 min, 60 min and 90 min. Only 80 °C overlaps direct Prusament characterization; overlap alone does not validate recovery or duration. Higher temperatures are not admitted by extrapolation. Geometry gaps remain candidates. Production case registries contain no approved cases because physical validity and production convergence are incomplete. No representative dry run or campaign was executed.

3.15. Sensitivity and uncertainty

No surrogate or cross-validation score exists. Future fitting needs traceable production cases and held-out assessment grouped by physical condition; preprocessing and tuning belong inside training folds. FREE is a distinct boundary condition, and contact transitions require sufficient sampling. Training agreement alone cannot establish prediction quality.

Deterministic process variation is distinct from physical-input uncertainty. No admitted probability distribution exists for modulus, expansion, conductivity, heat capacity, irreversible strain, convection, friction or fixture expansion. Morris, Sobol and Latin-hypercube/Monte Carlo methods remain conditional on supported domains, joint inputs and estimator convergence; none was executed. Verification discretization, surrogate error and physical-model discrepancy cannot substitute for propagated physical uncertainty.

3.16. Optimization and confirmation

Only actual outputs could support minimizing released warpage, dimensional error and defined residual stress. Contact limits, cycle time and annealing-benefit criteria need justification. Tensile strength is excluded because no validated strength/fracture model exists. Equal, dimensional-fidelity and stress/contact priorities have no assigned scales or thresholds; desirability is not used.

A future sampled Pareto set remains conditional on feasibility, resolution and uncertainty. A recommendation, two feasible neighbors and a thermally matched FREE case would require frozen predictions followed by new refined ANSYS runs. Identical oven programs do not establish thermal matching. No selection, processing window or confirmation error exists.

4. Results

4.1. Verification

All solver results below are numerical reference responses, not production PLA coupon predictions or independent physical validation. Across nine preselected center-temperature samples, the plane-wall maximum absolute error is 0.005180700 °C, maximum kelvin-based error 0.001700533% and maximum excursion-relative error 0.157660518%, meeting both limits. This is not a maximum over all field locations or solution times. Retained decimal digits support reproduction of exported numbers and do not imply physical measurement accuracy. Four preceding attempts remain archived: extraction failure, two rejected time-step comparisons and a file-mapping termination.

Six structural/contact cases pass all 114 comparisons. Maximum axial-displacement and gap errors are 2.331×10^{-14} mm and 9.500×10^{-11} mm. Contact is open at 30 °C, closed at 80 °C and open after cooling. Free elastic cycling recovers original dimensions; the imposed-eigenstrain case retains its prescribed contraction. These are implementation findings, not PLA shrinkage measurements.

Table 1. Structural reference checks: selected genuine outputs

Case and state	ANSYS value	Analytical value
Free, 80 °C: axial displacement	0.006000000 mm	0.006000000 mm
Fixed, 80 °C: axial stress	−1.200000048 MPa	−1.200000000 MPa
Fixed, 80 °C: left reaction	1.200000000 N	1.200000000 N
Free, cooled: axial displacement	0 mm	0 mm
Retained strain, cooled free: displacement	−0.010000000 mm	−0.010000000 mm
Retained strain, cooled fixed: stress	2.000000000 MPa	2.000000000 MPa
Retained strain, released: displacement	−0.010000000 mm	−0.010000000 mm
Contact, 80 °C: axial displacement	0.002000000 mm	0.002000000 mm
Contact, 80 °C: signed element axial force	−0.800000012 N	−0.800000000 N
Contact, cooled: open separation	0.002000000095 mm	0.002000000000 mm

The negative contact-force entry follows the CONTA178 axial output convention; its magnitude is 0.800000012 N. Compressive surface pressure in Eq. (14) is positive, and is a different quantity.

The Prony reference has maximum stress error 3.8271×10^{-6} MPa, maximum relative stress error 0.019740% and maximum reaction error 3.8280×10^{-6} N. The earlier 0.01 s ramp attempt failed the fixed criterion and remains archived. Negative stress after unloading is held-strain memory. Detailed thermal and Prony comparisons are in Supplementary Tables S5 and S6.

4.2. Mesh convergence

Initial medium-to-fine structural changes were 9.9345% for warpage, 7.9763% for displacement and 29.8796% for stress, requiring further refinement. The 90×12 grid then passed against 120×16 . The 32-element thermal grid passed against 64, and the normal-Lagrange 40-element interface against 60.

Table 2. Selected verification meshes and confirmation changes

Configuration	Selected / confirmation grid	Confirmation changes, δ_q
Two-layer Prony cantilever	90×12 / 120×16 elements	$W_{\max}$ 0.9327%; displacement 0.5821%; $\sigma_{\text{res},95}$ 2.0266%
Plane-wall thermal field	32 / 64 elements through thickness	300 s lag 0.0009423%; secant gradient 0.0060073%
Normal-Lagrange contact	40 / 60 interface elements	Mean pressure 0.12698%; peak pressure 0.36127%; reaction 0.13831%

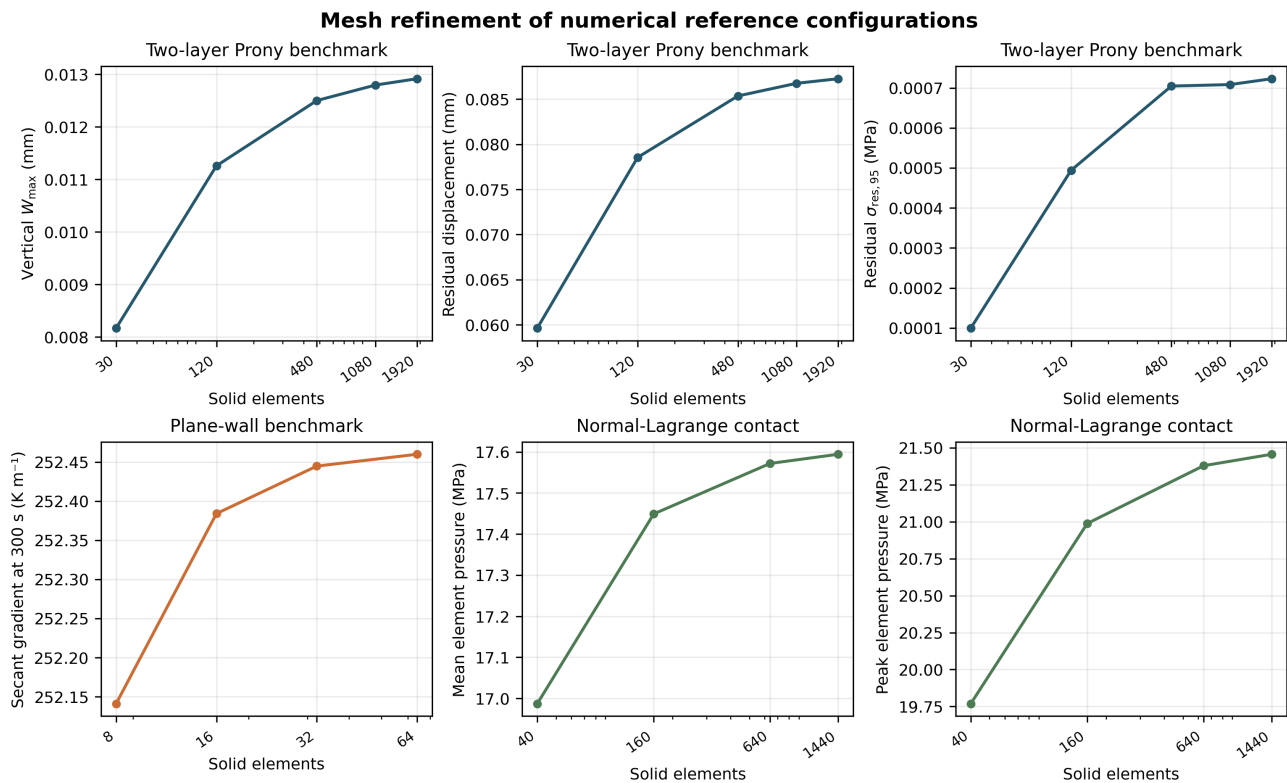


Figure 2. Genuine MAPDL mesh-refinement responses for numerical structural, thermal and surface-contact configurations. Every point traces to an archived solver run; no point is a production annealing prediction.

These selections apply only to the benchmark configurations. Twenty-eight unique runs populate the convergence tables; all 43 technical directories, including 15 excluded attempts, remain archived. No production mesh or time step is selected.

4.3. Time-step convergence

Table 3. Selected verification time increments and confirmation changes

Configuration	Selected / confirmation increments	Confirmation changes, δ_q
Plane-wall thermal history	0.25 s / 0.125 s	Lag 0.005285%; secant gradient 0.005369%; profile RMS excursion 0.002811%; profile RMSE 0.00202088 °C
Two-layer structural history	0.01/0.1 s / 0.005/0.05 s ramp/hold	$W_{\max}$ 0.003094%; displacement 0.003213%; $\sigma_{\text{res},95}$ 0.13636%

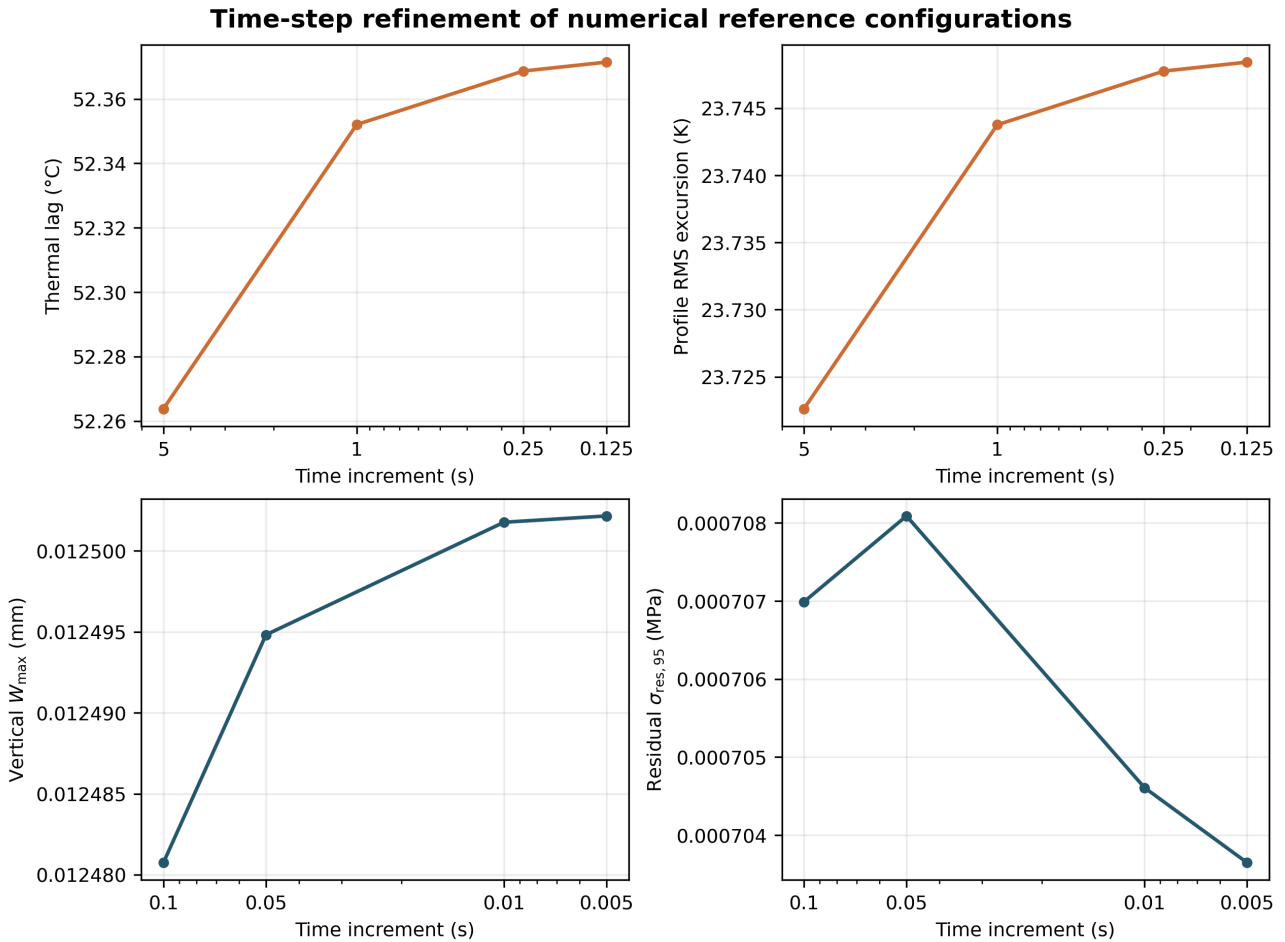


Figure 3. Thermal and structural time-step verification. The thermal profile comparison gives RMSE = 0.00202088 °C. Confirmation changes satisfy the predeclared benchmark limits. Structural abscissae are ramp increments; hold increments are ten times larger. Profile RMS excursion is distinct from pairwise profile RMSE.

Contact-control variation is distinct from discretization. At $F_{\text{KN}} = 10$, an edge element reports zero average pressure and the positive-pressure engagement screen fails despite convergence. The data do not resolve closure at every integration point. Penalty and augmented-Lagrange solutions coincide only for the declared monotonic frictionless benchmark. Under the same augmented-Lagrange $F_{\text{KN}} = 1$ control, increasing μ from 0 to 0.3 changes mean pressure from 17.5368 to 20.0334 MPa and peak pressure from 24.4284 to 51.8788 MPa (Supplementary Table S7). This is numerical-fixture sensitivity, not a measured PLA–steel friction effect.

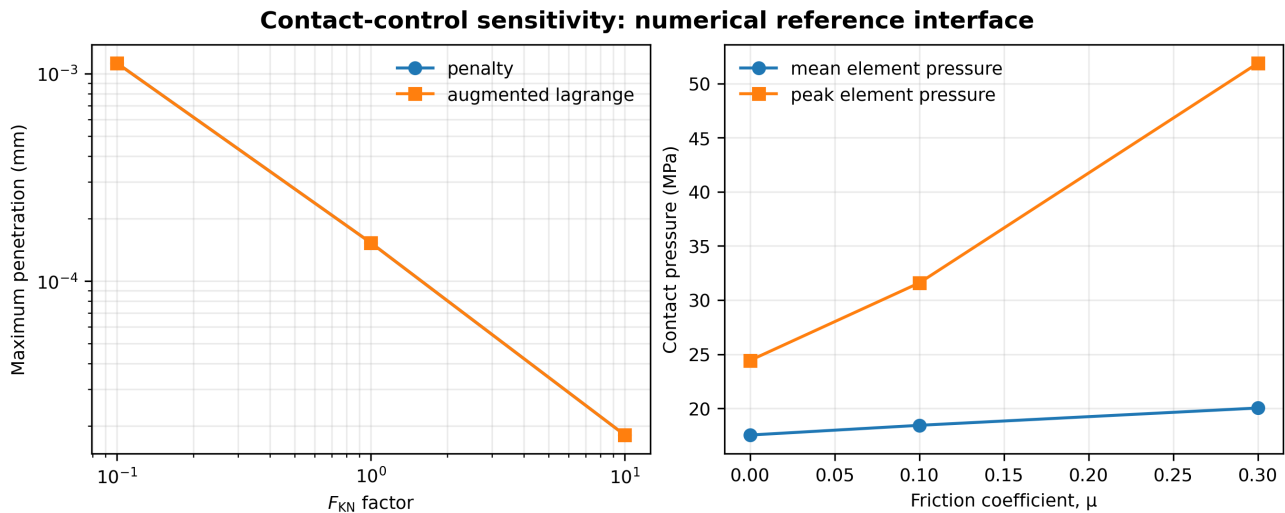


Figure 4. Contact-control responses in the numerical fixture. Penalty stiffness and friction are declared perturbations rather than sourced production interface parameters. Overlapping penalty/augmented curves are retained; the $F_{KN} = 10$ points fail the positive-pressure engagement screen. Algorithm differences must be retained when comparing cases.

4.4. Validation

All six source-specific cases were rejected before ANSYS launch because compatible material, initial-state, thermal/support and implementation inputs are incomplete. All 18 reserved means have blank predictions/errors. No recalibration was performed. This is missing-input rejection, not poor prediction–measurement agreement or physical validation. Imposing observed shrinkage as an input would destroy independence. Production sweeps remain blocked.

4.5. Thermal response

No production part/fixture temperature history or gradient was computed. Plane-wall agreement establishes only its constant-property convection problem.

4.6. Free annealing

No production FREE solution exists. Free expansion and imposed-eigenstrain patch tests do not replace it.

4.7. Fixture-clearance effect

No matched FREE–GAP pair exists. Geometry construction and node-stop closure do not quantify released dimensional preservation.

4.8. Temperature effect

No production ranking is available; 95 °C and 110 °C remain outside the constitutive reference envelope.

4.9. Holding-time effect

No 30 min, 60 min or 90 min production hold was solved. Isothermal reference relaxation cannot establish their comparative performance.

4.10. Coupled response

Thermal–contact–recovery interactions remain uncomputed because the non-isothermal model, initial state and interfaces are incomplete.

4.11. Dimensional stability

No signed production dimensional change, combined error or suppression percentage exists. Reserved external means remain literature observations.

4.12. Warpage

Vertical fitted-line warpage is a two-layer cantilever response, not released three-dimensional coupon best-fit-plane warpage.

4.13. Residual stress

Benchmark interior stress and held-strain Prony memory do not quantify production residual stress after cooling and release.

4.14. Contact pressure

Only reference pressures exist. No production pressure field, contact area, onset temperature or fixture reaction was predicted.

4.15. Sensitivity

No production global sensitivity indices or process-interaction estimates exist. Contact-control perturbations are verification cases.

4.16. Uncertainty

No propagated interval exists. Supplier intervals, method brackets and reference mesh differences are not confidence limits on a production prediction.

4.17. Optimization

No Pareto front, nondominated processing condition, preference ranking or robust window was computed. Empty registries record unavailable analysis, not established infeasibility.

4.18. Confirmation

No recommendation, neighbor or matched FREE point was selected. No new confirmation run or prediction error exists; reference refinements do not confirm an optimum.

5. Discussion

The findings support implementation-level confidence for specified thermal, elastic, imposed-eigenstrain, isothermal viscoelastic and contact problems. They also demonstrate why solver completion is insufficient: early time increments missed criteria, contact-control placement required correction, and one converged case failed its element-pressure engagement screen. Preserving these attempts prevents successful terminal status from being mistaken for adequacy.

The evidence does not yet answer the clearance–distortion–stress question. Initial printed state and irreversible strain are central: reversible expansion does not supply an absent annealing mechanism, and finite-time delayed deformation is not necessarily irreversible. Combining recovered pre-strain, initial stress and crystallization distortion without identification could count the same response twice. The implemented relaxation clock likewise cannot be multiplied by an unrelated modulus-reduction curve without justification.

Fixture engagement can alter thermal exposure and restraint. Attribution requires actual specimen/plate histories, resolved gap and contact status, pressure and penetration. Lower hot displacement need not imply lower released warpage. Fixture expansion, release relaxation and response state remain essential. No production mechanism, inconvenient trend or beneficial clearance is asserted because those fields do not exist.

Validation is blocked by compatibility, not demonstrated failure of finite elements to reproduce PLA. Prusament DMA parameters cannot supply Ultimaker recovery, furnace setpoint cannot reconstruct unknown cooling, and geometric agreement alone cannot validate stress or pressure. Reserved observations

expose missing links before fitting. Verification convergence and surrogate agreement cannot replace independent physical validation.

Limitations include small-strain isotropy, absent thermal/recovery functions, unknown fixture expansion convention and unresolved boundaries. Orthotropy, crystallization and finite rotations require evidence and implementation rather than decorative equations. Sand/salt remains literature context; no discrete-element model or new physical experiment is claimed. The contribution is reproducible verification, not manufacturing qualification, strength enhancement or optimized treatment. The intended ideal centered fixture omits gravity, actual spacer expansion and assembly tolerances; its clearances are not demonstrated achievable shop-floor settings. No productivity, cost, energy or service-life improvement has been calculated. Normal-Lagrange refinement establishes stability of the sampled reference pressures over the tested meshes, not a unique physical contact solution or convergence of edge singularities.

6. Conclusions

Seven quantitative findings are supported by archived numerical reference cases:

1. Across nine sampled center temperatures, the plane-wall maximum absolute temperature error is 0.005180700 °C and maximum excursion-relative error 0.157660518%, below the declared 0.05 °C and 0.25% limits.
2. Six structural/contact cases pass 114 comparisons, with maximum displacement and gap errors of 2.331×10^{-14} mm and 9.500×10^{-11} mm.
3. The 23-branch isothermal Prony reference has maximum stress error 3.8271×10^{-6} MPa and relative stress error 0.019740%. This checks the 65 °C adapter, not non-isothermal recovery.
4. Structural refinement from 90×12 to 120×16 elements changes vertical fitted-line warpage, displacement and interior stress by 0.9327%, 0.5821% and 2.0266%.
5. Thermal refinement from 32 to 64 through-thickness elements changes 300 s lag by 0.0009423% and secant gradient by 0.0060073%.
6. The 0.25 s versus 0.125 s thermal comparison changes lag by 0.005285% and gives profile RMSE 0.00202088 °C. Structural ramp/hold refinement from 0.01/0.1 s to 0.005/0.05 s changes warpage by 0.003094% and stress by 0.13636%.
7. Normal-Lagrange refinement from 40 to 60 interface elements changes mean pressure, peak pressure and reaction by 0.12698%, 0.36127% and 0.13831%, with zero reported penetration.

These results apply to numerical fixtures and do not establish production mesh adequacy, annealing benefit, physical predictive validity or optimum. The full computational annealing study remains scientifically incomplete. Compatible material/boundary evidence, non-isothermal implementation and independent validation must precede production inference, uncertainty propagation and processing recommendations.

Data availability

The [FDM-Annealing repository](#) contains solver evidence, failed/superseded attempts, case definitions, extraction tables and SHA-256 manifests. Literature-derived observations retain source locations and calibration/validation roles. Production and optimization registries are header-only; there are no withheld production results.

Code availability

Material relations, MAPDL deck generation, analytical references, metric functions, checks and the PDF builder are maintained in the repository. The manuscript equation map identifies actual implementations and their scope. Ansys is proprietary software; the recorded installation defines the solver environment. No claim is made that every model runs without it.

Declarations

No new physical specimens, human participants or animal studies form part of this computational work. Names and affiliation are retained from the supplied base manuscript. Author contributions, corresponding-author details, funding and competing-interest declarations require author confirmation before submission; absence is not inferred. This is not a completed production-annealing or optimization article ready for submission.

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Supplementary information

S1. Literature comparison and reproducibility tables

Literature values below are source evidence, not new experiments or solver findings. Full matrices, formulation flags and exact locators remain in the repository. Reference identities were checked against the bibliography. The focused September 2026 audit adds an ABS clearance precedent [41] without treating it as PLA material evidence.

Table S1. Closest competing studies and implications for scope

Study	Established overlap	Distinction still requiring evidence
Bute et al. [5]	Irreversible strain and ANSYS printing stress	Predictive annealing-clearance/contact relation
Chapuis et al. [8]	Viscoelastic pre-strain recovery FE	Fixture-controlled dimensional preservation
Issabayeva and Shishkovsky [9]	ANSYS, DMA, Prony and thermal response	Released fixture geometry and stress
Wijnen et al. [6]	Warpage model and annealing dimensions	Quantified annealing restraint
Trofimov et al. [7]	Thermal/deformation prediction and comparison	Post-print annealing contact cycle
Lluch-Cerezo et al. [11]	Powder-supported dimensional control	Explicit clearance and contact mechanics
Wijnbergen et al. [12]	Tough-PLA annealing media	Compatible grade and plate-gap prediction
Chiscop et al. [24]	Encapsulation across geometries	Contact-clearance response under release
De Assis et al. [13]	Geometry/mechanics and thermal conditioning	Transient fixture-contact model
Ben Amor and Souissi [14]	Annealed PLA, FE and optimization	Released dimensions–warpage–stress decision

Table S2. Prusament PLA generalized-Maxwell coefficients at $T_g = 65\text{ }^{\circ}\text{C}$ from Chapuis et al. [8, supplementary Table B.2]

i	k_i (MPa)	τ_i (s)
1	6.228	1.176×10^{-14}
2	1.806	1.228×10^{-14}
3	7.849	2.878×10^{-14}
4	12.751	1.701×10^{-12}
5	20.857	9.178×10^{-12}
6	31.104	6.034×10^{-11}
7	45.566	6.187×10^{-9}
8	64.036	1.102×10^{-7}
9	92.484	1.047×10^{-6}
10	135.916	6.903×10^{-6}
11	113.488	2.582×10^{-5}
12	128.266	5.363×10^{-5}
13	173.470	1.838×10^{-4}
14	93.393	4.386×10^{-4}
15	163.804	7.577×10^{-4}
16	134.311	1.795×10^{-3}
17	119.266	2.292×10^{-3}
18	100.596	6.576×10^{-3}
19	80.534	9.920×10^{-3}
20	61.624	3.290×10^{-2}
21	44.508	5.374×10^{-2}
22	30.529	6.586×10^{-2}
23	18.610	3.503×10^1

Table S3. Candidate AISI 304 fixture properties from Meng et al. [38, Table 1]

Property	20 °C	100 °C / 200 °C
ρ	7910 kg m ⁻³	7876 / 7840 kg m ⁻³
c_p	456 J kg ⁻¹ K ⁻¹	494 / 532 J kg ⁻¹ K ⁻¹
k	16.2 W m ⁻¹ K ⁻¹	16.6 / 17.45 W m ⁻¹ K ⁻¹
α	15.5 μ m m ⁻¹ K ⁻¹	16.3 / 16.7 μ m m ⁻¹ K ⁻¹
E	200 GPa	191.4 / 183.5 GPa
ν	0.290	0.285 / 0.289

Table S4. Reserved experimental dimensional means from Lluch-Cerezo et al. [11, Table 5], without mould

T (°C)	Length change (%)	Width change (%)	Height change (%)
63	-0.13	-0.06	0.00
75	-1.60	-0.11	2.74
86	-2.30	-0.16	2.62
98	-2.88	0.15	2.60
109	-3.05	0.30	3.94
132	-3.58	0.18	3.69

Table S5. Genuine ANSYS–analytical center-temperature comparison for the plane-wall verification

Time (s)	ANSYS / analytical (°C)	Absolute / excursion-relative error
150	21.892109254 / 21.889130841	0.002978413 °C / 0.157660518%
300	27.631062216 / 27.625947421	0.005114795 °C / 0.067070946%
450	34.805348857 / 34.801718199	0.003630658 °C / 0.024528625%
600	41.003870402 / 41.001387994	0.002482407 °C / 0.011820207%
900	50.967098637 / 50.966218772	0.000879865 °C / 0.002841371%
1050	53.056900023 / 53.059537028	0.002637005 °C / 0.007976533%
1200	50.753732030 / 50.758912731	0.005180700 °C / 0.016842924%
1500	42.903452036 / 42.906520281	0.003068244 °C / 0.013394633%
1800	37.051786082 / 37.053501761	0.001715679 °C / 0.010060566%

Table S6. Isothermal Prony finite-ramp stress verification at 65 °C

Time (s)	ANSYS stress (MPa)	Analytical stress (MPa)	Absolute error (MPa)
1	0.0375536568	0.0375574835	0.0000038266
11	0.0243886374	0.0243886379	0.0000000005
31	0.0183896348	0.0183896352	0.0000000005
32	-0.0193833038	-0.0193871309	0.0000038271
62	-0.0045757745	-0.0045757747	0.0000000001

Table S7. Contact-control sensitivity on the 40-element interface

Control	Maximum penetration (mm)	Mean / peak pressure (MPa)
Penalty or augmented, $F_{KN} = 0.1$	1.12141×10^{-3}	16.9989 / 17.9426
Penalty or augmented, $F_{KN} = 1$	1.52677×10^{-4}	17.5368 / 24.4284
Penalty or augmented, $F_{KN} = 10$	1.80923×10^{-5}	17.6242 / 28.9476
Normal Lagrange, $\mu = 0$	0	17.5721 / 21.3788
Augmented, $F_{KN} = 1$, $\mu = 0.1$	1.97427×10^{-4}	18.4343 / 31.5883
Augmented, $F_{KN} = 1$, $\mu = 0.3$	3.24243×10^{-4}	20.0334 / 51.8788

S2. Evidence and reproduction boundaries

The original literature matrix contains 34 journal studies; the separate audit addendum records the additional ABS clearance study [41]. The base paper is immutable. Property records preserve units, temperatures, formulation, exact source location and use restrictions. Full convergence CSVs contain 39 mesh, 24 time-step and nine contact-control rows; these are response records, not independent experiments. All 43 technical convergence directories remain archived, including 15 excluded attempts.

Verification constants, load schedules, relaxation mismatch and clearance screens are design choices. The assembly construction is not a solved annealing model. The validation archive retains 45 published observations: 18 reserved means, 24 context records and three quarantined maxima. Prediction/error fields remain blank. Empty registries do not mean zero physical response.

Each run preserves input, script, command, raw output and extraction hashes. Corrections create new derived records or attempts; raw evidence is not overwritten. The supplementary files support numerical reproduction and do not supply missing physical validation.